

VERA Mic Harness — Filtered Power, Shielded Signal
v1.24.2 • US Letter

Build Notes

GROUNDING (no Pi pin 6 available):

- Amp-side ground: I²C-header GND pin only. ALL caps + MAX GND tie here.
- Receiver-side ground: 3.5 mm TRS sleeve only. Twisted-pair return + foil drain go here.
- Copper foil shield FLOATS at the amp end. Soldered only to the TRS sleeve via drain wire.
- The two ground points are connected through the Pi's internal GND plane (via USB ground of CM108 ↔ Pi USB shell ↔ I²C-header GND). This forms ONE long ground path, not a loop.

COMPONENT PLACEMENT:

- Mount C1/C2/C3 within 5 mm of MAX4466 VDD pin. Lead length kills HF filtering.
- Ferrite + 10 Ω + caps form a π -filter to remove Pi-rail switching ripple.
- R_{lp} + C_{lp} (1 k Ω + 4.7 nF C0G) cut everything above 34 kHz → blocks RF.

CABLE ROUTING:

- Twist signal and return ≥ 6 turns/inch for common-mode rejection.
- Helical-wrap copper foil; $\geq 50\%$ overlap per turn.
- Keep harness ≥ 25 mm away from servo PWM, fan PWM, USB and AC wires.
- If crossing any of those wires is unavoidable, cross at 90°.

VERIFY:

arecord -D plughw:2,0 -d 5 -f cd /tmp/silence.wav
sox /tmp/silence.wav -n stat → RMS < 0.001, Max amp < 0.01

